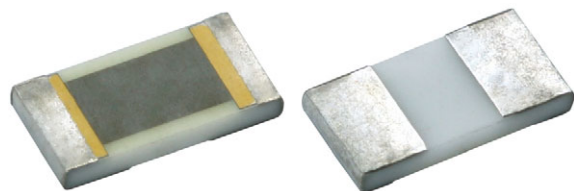


High Power Thin Film Wraparound Chip Resistor



Resistor patterns and general appearance may vary.

LINKS TO ADDITIONAL RESOURCES


[3D Models](#)

[Infographics](#)

[Product Page](#)

[Packages](#)

[Footprints](#)

[Order Samples](#)

PHP series chip resistors are designed with enlarged backside terminations to reduce the thermal resistance between the topside resistor layer and the solder joint on the end users circuit board.

Actual power handling capability is limited by the end user mounting process. As with any high power chip resistor the ability to remove the generated heat is critical to the overall performance of the device.



ATTENTION!
Observe Precautions for
Handling Electrostatic Sensitive Devices!

FEATURES

- High purity ceramic substrate
- Power rating to 2.5 W
- Resistance range 10 Ω to 30.1 k Ω
- Resistor tolerance to ± 0.1 %
- TCR to ± 25 ppm/ $^{\circ}$ C
- Flame resistant UL 94 V-0
- Material categorization:
for definitions of compliance please see
www.vishay.com/doc?99912



Pb-free

RoHS*

Available

HALOGEN
FREE

Available

GREEN

(5-2008)

Available

Note

* This datasheet provides information about parts that are RoHS-compliant and / or parts that are non RoHS-compliant. For example, parts with lead (Pb) terminations are not RoHS-compliant. Please see the information / tables in this datasheet for details

APPLICATIONS

- Power supplies
- Power switching
- Braking system
- Test and measurement equipment
- Motor deflection circuits

TYPICAL PERFORMANCE

	ABSOLUTE
TCR	25
TOL.	0.1

STANDARD ELECTRICAL SPECIFICATIONS

TEST	SPECIFICATIONS	CONDITIONS
Material	Nichrome	-
Resistance range	10 Ω to 30.1 k Ω	-
TCR: absolute	25 ppm/ $^{\circ}$ C, 50 ppm/ $^{\circ}$ C (standard) and, 100 ppm/ $^{\circ}$ C	-55 $^{\circ}$ C to +125 $^{\circ}$ C
Tolerance: absolute	0.1 %, 0.5 %, 1.0 % and, 5.0 %	+25 $^{\circ}$ C
Power rating: resistor	0.375 W to 2.5 W ⁽¹⁾	Maximum at +70 $^{\circ}$ C
Stability: absolute	ΔR 0.1 %	2000 h at +70 $^{\circ}$ C
Stability: ratio	Not applicable	-
Voltage coefficient	< 0.1 ppm/V	-
Working voltage	75 V to 200 V	-
Operating temperature range	-55 $^{\circ}$ C to +155 $^{\circ}$ C	-
Storage temperature range	-55 $^{\circ}$ C to +155 $^{\circ}$ C	-
Noise	< -30 dB	-
Shelf life stability: absolute	± 0.01 %	1 year at +25 $^{\circ}$ C

COMPONENT RATINGS

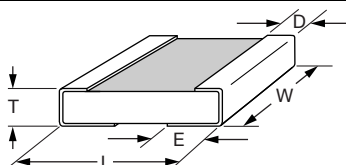
CASE SIZE	POWER RATING (mW)	WORKING VOLTAGE (V)	RESISTANCE RANGE (Ω)
0603	375 ⁽¹⁾	75	10 to 30.1K
0805	625 ⁽¹⁾	100	10 to 30.1K
1206	1000 ⁽¹⁾	200	10 to 30.1K
2512	2500 ⁽¹⁾	200	10 to 30.1K

Note

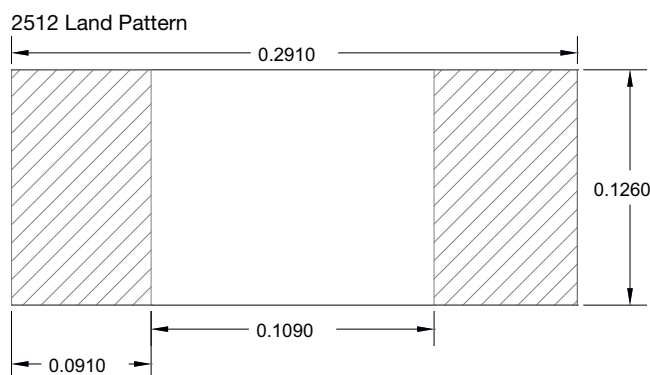
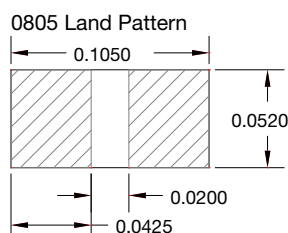
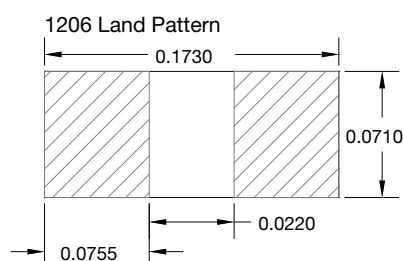
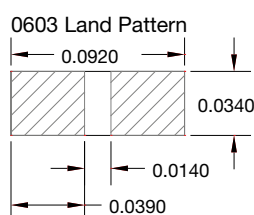
⁽¹⁾ Dependent on component mounting by user

ENVIRONMENTAL TESTS (Vishay Performance vs. MIL-PRF-55342 Requirements)

ENVIRONMENTAL TEST	LIMITS MIL-PRF-55342 CHARACTERISTIC "E"	TYPICAL VISHAY PERFORMANCE
Resistance temperature characteristic	$\pm 25 \text{ ppm/}^{\circ}\text{C}$	$\pm 15 \text{ ppm/}^{\circ}\text{C}$
Maximum ambient temperature at rated wattage	$+70^{\circ}\text{C}$	$+70^{\circ}\text{C}$
Maximum ambient temperature at power derating	$+150^{\circ}\text{C}$	$+150^{\circ}\text{C}$
Thermal shock	$\pm 0.1 \%$	$\pm 0.04 \%$
Low temperature operation	$\pm 0.1 \%$	$\pm 0.001 \%$
Short time overload	$\pm 0.1 \%$	$\pm 0.003 \%$
High temperature exposure	$\pm 0.1 \%$	$\pm 0.030 \%$
Resistance to soldering heat	$\pm 0.2 \%$	$\pm 0.007 \%$
Moisture resistance	$\pm 0.2 \%$	$\pm 0.002 \%$
Life at $+70^{\circ}\text{C}$ for 2000 h	$\pm 0.5 \%$	$\pm 0.100 \%$

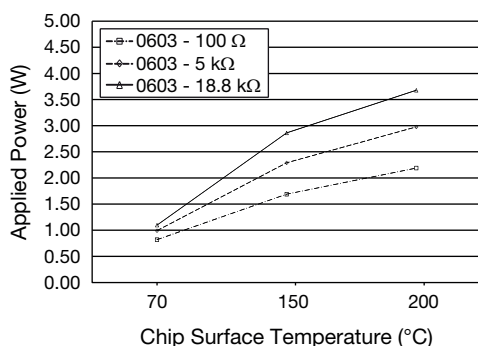
DIMENSIONS in inches


CASE SIZE	LENGTH	WIDTH W (± 0.005)	THICKNESS MIN./MAX.	TOP PAD D (± 0.005)	BOTTOM PAD E (± 0.005)
0603	0.064 ± 0.006	0.032	0.020 max.	0.012	0.021
0805	0.080 ± 0.006	0.050	0.015/0.033	0.016	0.025
1206	0.126 ± 0.008	0.063	0.015/0.033	$0.020 + 0.005/- 0.010$	0.040
2512	$0.259 + 0.009/- 0.015$	0.124	0.015/0.033	0.02	0.050

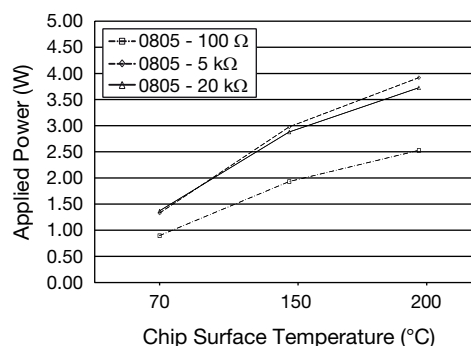
LAND PATTERN DIMENSIONS in inches


**STANDARD MATERIAL SPECIFICATIONS**

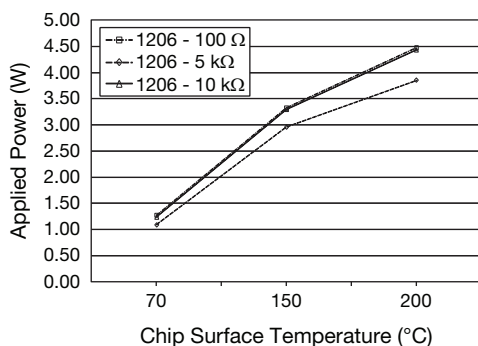
Resistive element	Nichrome
Substrate material	Alumina (Al ₂ O ₃)
Terminations (tin / lead)	Tin / lead solder over nickel barrier
Terminations (lead (Pb)-free)	Tin / silver / copper (Sn96.5Ag3.0Cu0.5) solder over nickel barrier

PHP CHIP TEMP. VS. APPLIED POWER**Note**

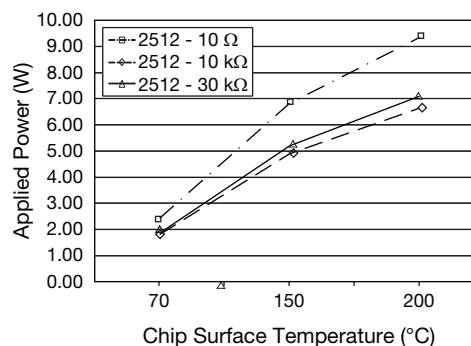
- Chip surface temperature measured using FLIR SC645 thermal imaging system with an approximate test card surface temperature of 85 °C

PHP CHIP TEMP. VS. APPLIED POWER**Note**

- Chip surface temperature measured using FLIR SC645 thermal imaging system with an approximate test card surface temperature of 85 °C

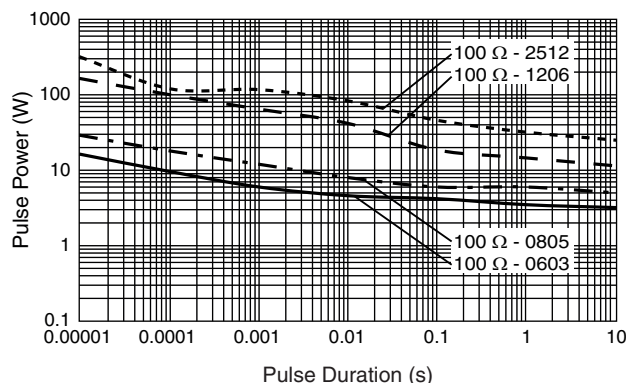
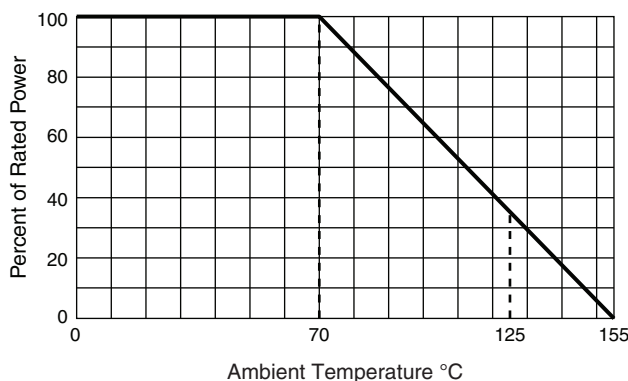
PHP CHIP TEMP. VS. APPLIED POWER**Notes**

- Chip surface temperature measured using FLIR A40 thermal imaging system with an approximate test card surface temperature of 25 °C
- Thermal imaging was conducted under ambient conditions resulting in a steady state test card surface temperature of 85 °C over the full range of power levels
- Thermal imaging and load life testing was conducted mounting one device to 2" x 3" test cards with 2.5 mil copper plating on both surfaces. Thermal vias on 120 mil centers were utilized for heat transfer between surfaces of the test card

PHP CHIP TEMP. VS. APPLIED POWER**Notes**

- Chip surface temperature measured using FLIR A40 thermal imaging system with an approximate test card surface temperature of 25 °C

Case Size	2512	2512	2512
Resistance Value	Up to 10 Ω	Up to 10 kΩ	Up to 30 kΩ
Temperature	Power (W)		
70	2.44	1.81	1.87
150	6.82	4.89	5.19
200	9.33	6.63	7.09

SINGLE PULSE CURVES

DERATING CURVE

GLOBAL PART NUMBER INFORMATION

P	H	P	0	1	2	0	6	E	1	0	0	2	B	B	T	1
GLOBAL MODEL	SUBSTRATE	CASE SIZE	TCR	RESISTANCE	TOLERANCE	TERMINATION	PACKAGING									
PHP	0 = alumina	0603 0805 1206 2512	E = ± 25 ppm/°C H = ± 50 ppm/°C K = ± 100 ppm/°C	The first 3 digits are significant figures and the last digit specifies the number of zeros to follow. “R” designates the decimal point. Example: 10R0 = 10 Ω 1000 = 100 Ω 1001 = 1 kΩ	B = ± 0.1 % D = ± 0.5 % F = ± 1.0 % G = ± 2.0 % J = ± 5.0 %	B = wraparound Sn/Pb solder w/nickel barrier S = wraparound lead (Pb)-free solder SAC-305 RoHS-compliant - e1	BS = BULK 100 min., 1 mult. WS = WAFFLE 100 min., 1 mult. WI = WAFFLE (item single lot date code) 100 min., 1 mult. TAPE AND REEL T1 = 1000 min., 1000 mult. T3 = 300 min., 300 mult. T5 = 500 min., 500 mult. TF = full reel TS = 100 min., 1 mult. TI = 100 min., 1 mult. (item single lot date code) TP = 100 min., 1 mult. (package unit single lot date)									



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